

RED ALGAE

Although some are microscopic organisms, the most familiar red algae are multicellular seaweeds, which are amongst the largest of the protocists. They have evolved into an extraordinary diversity of forms, ranging from rock-hugging encrustations that look more like lichens to bristly or leafy clumps.

The great majority of the 6,500 or so species of red algae are marine, but there are a few freshwater species. Unlike brown and green algae, they do not produce flagellated sperm and rely instead on water currents to transport male sex cells to the female organs. The cells then merge, but subsequent development is highly variable between species.

RED FOR SURVIVAL

Like land plants, most algae use green chlorophyll to trap the energy of sunlight and manufacture food in the process of photosynthesis. However, the cells of red algae also contain a red pigment, called phycoerythrin, which masks the green colour of the chlorophyll and gives them their red colour – and their name. This pigment allows them to continue photosynthesizing even at depths in the water where only a little blue light can penetrate. This allows red algae to live much deeper in the oceans than brown or green seaweeds, and some have been reported growing as deep as 200 m (650 ft).

Some red algae become calcified, forming encrustations on rocks or hard, upright, antler-like forms. And, in spite of their name, their pigments can make red algae appear olive or grey.

DOMAIN	EUKARYOTA
KINGDOM	PROTOCTISTA
DIVISION	RHODOPHYTA
FAMILIES	92
SPECIES	About 6,500

DEBATE

CLOSE RELATIVES?

Red algae were previously grouped with green algae, but it is now recognized that green algae are two separate and very diverse groups. Some species may be close to red algae, but others are closer to land plants. Red algae branched off the taxonomic tree before the land plants began to evolve.



DULSE

Palmaria palmata

In N. Atlantic coastal communities, this edible seaweed is a traditional source of protein and vitamins.



CORAL WEED

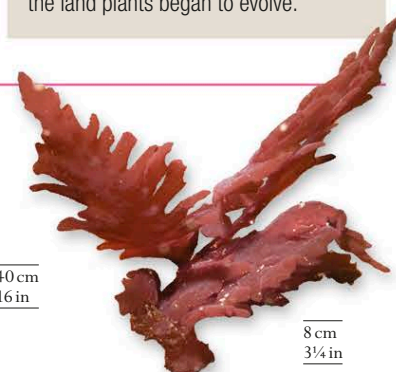
Corallina officinalis

This red seaweed is common in rock pools worldwide, forming branching, feather-like fronds.



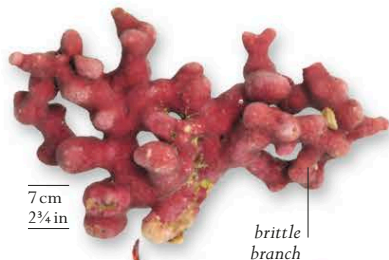
AGARDHIELLA SUBULATA

Originally from the western Atlantic, Caribbean, and Gulf of Mexico, this fleshy red alga has invaded some parts of Europe.



SCHMITZIA HISCOCKIANA

Found in areas swept almost bare by the tide, this red seaweed is fleshy and gelatinous, with flattened branches and finger-like projections.



MAERL

Phymatolithon calcareum

This species, an encrusting coralline alga of the British Isles, is commercially dredged and ground up as a calcium-rich soil additive.



BLACK CARRAGEEN

Furcellaria lumbricalis

This northern hemisphere seaweed has cylindrical, brown-black fronds that branch into masses of fleshy fingers.



EYELASH WEED

Calliblepharis ciliata

This northern hemisphere seaweed has flat fronds, fringed with numerous "branchlets".



GRACILARIA FOLIIFERA

Rising as slender dull purple stems, this red seaweed branches sparsely to give strap-like fronds. It grows abundantly in shallow lagoons worldwide.



GRACILARIA BURSA-PASTORIS

A long, thin red seaweed with forked or alternating branches, this species is found from southern England to the Pacific and the Caribbean.

tubular frond



fan-like branch



IRISH MOSS

Chondrus crispus

Also known as carrageen moss, this seaweed of the British Isles is important as a source of carrageen, which is a gelling agent.